



LASU Certificate Generator

OOP Design Patterns · C++17 · Winsock HTTP Server · Web UI

CSC 226 — Object-Oriented Programming I — Lagos State University — July 2026

Strategy

TemplateMethod

Factory

Inheritance

Polymorphism

RAII

std::map

Multithreading

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Overview & Architecture

Desktop webapp: generate, manage, and print HTML certificates

3 CertificateTypes

Excellence, Completion, Participation

3 Template Styles

Formal(serif),Modern(gradient), Minimalist (mono)

CSV Batch Import

File path or paste · 10 sample students

Search, Sort & Stats

Name,course,GPA · Per-course & grade distro



Data flow: Browser→HTTP(JSON) → HttpServer → CertificateGenerator → Certificate + Template → HTML file

Inheritance & Encapsulation

Person (abstract) → Student (concrete) · All data private

```
class Person { // Abstractbase
protected:
    std::string name_;
public:
    virtual ~Person() = default;
    virtual std::string
        getDescription()const= 0; // pure virtual
};

class Student : public Person { // is-a Person
    std::string course_, grade_;
public:
    double gradeToPoints() const;
    // 'A' ->4.0 'B+' ->3.3 '93' ->3.72
    static bool isValidGrade(const&);
    std::string getDescription()
        const override;
};
```

```
// Encapsulation in Generator
class CertificateGenerator {
    std::map<std::string, Student>
        students_; // private
    mutable std::mutex mutex_; // private
public:
    void addStudent(const&); // validated
    bool removeStudent(const&);
    Student* findStudent(const&);
    void loadFromCsv(const&);
};
```

- **Inheritance:** Student is-a Person — inherits name_, adds course/grade
- **Encapsulation:** All data private — addStudent() validates before storing, throws on bad data

Polymorphism & Abstract Base Classes

3 ABCs, 7 pure virtuals, runtime vtable dispatch

```
class Certificate{           // Abstractbase
public:
    virtual std::string
        generateHTML()const;    // Template Method
    virtual std::string
        getTypeName()const= 0; // pure virtual
};

class CertOfExcellence : public Certificate {
    std::string getTypeName()constoverride
    { return"Excellence"; }
};

class CertOfCompletion : public Certificate {
    std::string getTypeName()constoverride
    { return"Completion"; }
};
```

- **3 ABCs:** Person, CertificateTemplate, Certificate
- **7 pure virtual methods** total
- Runtime dispatch via vtable
- Client code works with base pointers

```
// Factory return subclass at runtime
auto tpl = makeTemplate(style);
auto cert = makeCertificate(
    type, s, std::move(tpl));
file << cert->generateHTML(); // polymorphic!
```

Strategy + Template Method

Strategy: swappable visual styles · Template Method: fixed skeleton

```
// STRATEGY: CertificateTemplate
class CertificateTemplate {
public:
    virtual string header(...)const= 0;
    virtual string styles()const= 0;
    virtual string body(...)const= 0;
};
class FormalTemplate : public CertTempl {
    // Double border, serif (Georgia)
};
class ModernTemplate : public CertTempl {
    // Gradient bg, rounded corners
};
class MinimalistTemplate : public CertTempl {
    // Monospace font, clean lines
};
```

```
// TEMPLATE METHOD: generateHTML()
string Certificate::generateHTML()const{
    string typeName = getTypeName(); // virtual
    string html;
    html += template_.header(typeName);
    html += template_.styles();
    html += template_.body(student_,
                           typeName, date);

    html += "</body></html>";
    return html;
}
// Subclass provides type name/desc
// Template provides all HTML/CSS
```

- **Strategy:** Open/Closed — new templates don't change Certificate | Style chosen at runtime
- **Template Method:** getTypeName() dispatches polymorphically | Zero duplication across types

Factory + Composition

makeTemplate/makeCertificate· has-a relationships throughout

```
//Factory: returns correct subclass
unique_ptr<CertificateTemplate>
makeTemplate(TemplateStyle style){
    switch (style) {
        case Formal:
            return mk<FormalTemplate>();
        case Modern:
            return mk<ModernTemplate>();
        case Minimalist:
            return mk<MinimalistTemplate>();
    }
}
```

```
// Composition: has-a
class Certificate {
    const Student& student_;
    unique_ptr<CertificateTemplate>
        template_; // has-a Template
};
```

- **Factory:** Specify an enum → get right polymorphic type
- No conditional logic at call site
- Add type: enum value + case + class
- **Composition:** has-a beats is-a
- Runtime flexibility (swap template)
- Looser coupling, easier testing
- HttpServer has-a Generator
- Generator has-a map of Students

RAII & STL Features

Automatic resource management · std::map, sort, find_if, lambdas, thread

```
// RAII: no manual cleanup
void addStudent(const Student& s) {
    std::lock_guard<mutex> lock(m_);
    students_[toLower(s.getName())] = s;
    // auto-unlocked at scope exit
}

// unique_ptr: auto-deleted polymorphic
auto cert = makeCertificate(type, s, t);
// cert + tmp auto-deleted when done
```

```
//STL: case-insensitive search
Student* findStudent(const string&n){
    auto it = find_if(begin(students_),
        end(students_),
        [&](const auto& p) {
            return toLower(p.first)
                == toLower(n);
        });
    return it != end(students_)
        ? &it->second : nullptr;
}
```

- **RAII:** unique_ptr, lock_guard, ofstream — zero manual delete/unlock/close
- **STL:** std::map (O(log n)), find_if+lambda, sort+lambda, thread, mutex, stringstream

HTTP Server & REST API

WinsockTCP · 12endpoints · Embedded SPA · Thread safety

```
// Accept loop: each conn on its own thread
while (true) {
    SOCKET client = accept(server,
        nullptr, nullptr);
    if (client == INVALID_SOCKET) break;
    std::thread(handleClient, client,
        std::ref(gen)).detach();
}
```

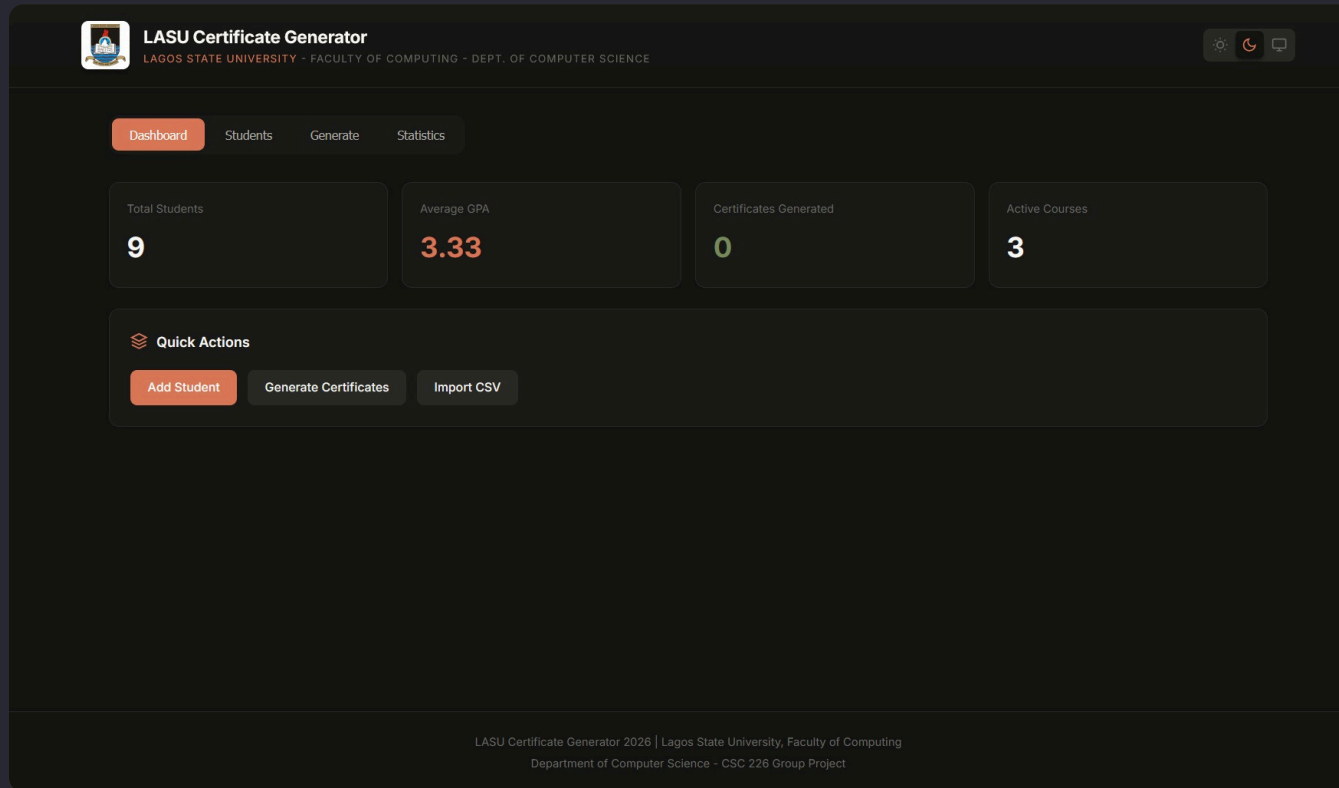
- All HTML/CSS/JS in C++ raw string literal
- Frontend uses fetch() — no reloads
- Theme toggle: Light/Dark/System

Method	Path	Action
GET	/	Serve Web UI
GET	/list	All students JSON
GET	/stats	Statistics JSON
POST	/add	Insert student
POST	/remove	Delete student
POST	/load	CSV import
POST	/generate	Create certificates
POST	/sort	Sort students

- std::mutex guards map (thread-safe)
- ValidationException → JSON error
- CSV: invalid lines skipped

Web UI Preview

Browser-based interface — Dashboard, Students, Generate, Statistics tabs



- **Dashboard:** Total students, avg GPA, cert count, active courses
- **Students:** Search, sort, add, remove — grade badges with colors
- **Generate:** Pick type + template style — one student or all
- **Statistics:** Course distribution, grade distribution
- **Theme toggle:** Light / Dark / Match System

All Concepts in Action

End-to-end: add student → generate certificate → HTML output

```
// 1. Add (encapsulation try { +exception)

    gen.addStudent(
        Student("Alice", "CSC 226", "A"));
} catch (const ValidationE& e) {
    cerr << e.what();
}

// 2. Create Template (factory + strategy) auto tmp1
= makeTemplate(Formal);

// 3. Create Certificate (factory + poly.) auto cert
= makeCertificate(
    Excellence, student, move(tmp1));
```

```
// 4. Generate HTML (templatemethod)
string html = cert->generateHTML();
//     getTypeName() → "Excellence"
//     FormalTemplate::header() + styles()
//     FormalTemplate::body()

// 5. Write file (RAII: ofstream) ofstream
file("output/Alice.html"); file << html; //
auto-closed

// 6. Serve via HTTP (thread + mutex) //
Browser GET /list returns JSON
```

Inheritance

Student : Person

Polymorphism

cert->generateHTML()

Encapsulation

private + getters

Abstract Base

3 ABCs, 7 virtuals

Strategy

FormalTemplate CSS

Template Method

generateHTML()

Factory

makeTemplate/Cert

Composition

has-a Template

RAII

unique_ptr/lock

STL

map/sort/thread

Exception

try/catch validate

Thread Safety

mutex guard

Thank You

Lagos State University—Department of Computer Science — CSC 226 — Group 6

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```
g++ -std=c++17 -I include main.cpp src/*.cpp -lws2_32 && ./certificate_generator
```

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